

Example A:

Out of 100 students in a school, 60 students play cricket and 52 students play football. There are 5 students who play neither of these games.

- 1) How many students play both cricket and football?**
- 2) How many students play cricket only?**
- 3) How many students play only football?**
- 4) How many students play either cricket or football but not both?**
- 5) How many students play exactly one game?**
- 6) How many students play cricket or football?**
- 7) How many students don't play football?**

I



100 students

$$\underline{\underline{95}} + \underline{\underline{17}} = 112$$

60 → Cricket
52 → Football
5 → Don't play anything

112

Example A:

$\boxed{08} \rightarrow \text{Inclusive / Exclusive}$
 $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$

Out of 100 students in a school, 60 students play cricket and 52 students play football. There are 5 students who play neither of these games.

1) How many students play both cricket and football?

2) How many students play cricket only? $\rightarrow 43$

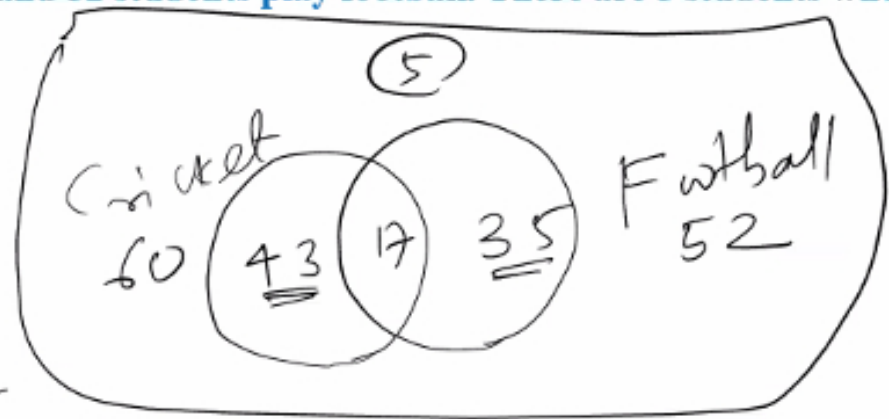
3) How many students play only football? $\rightarrow 35$

4) How many students play either cricket or football but not both? $\rightarrow 78$ $\left(\begin{array}{l} 43 + 35 \\ 95 - 17 \end{array} \right)$

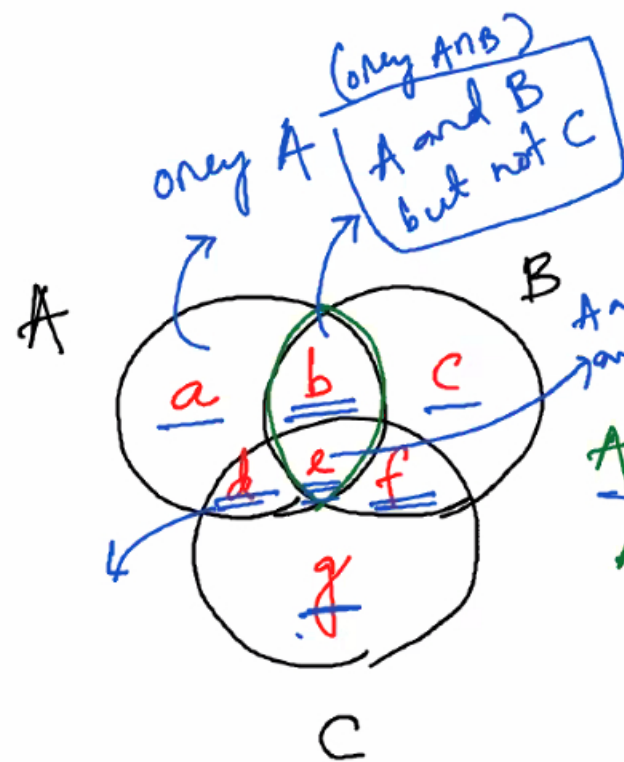
5) How many students play exactly one game?

6) How many students play cricket or football?

7) How many students don't play football?



$\rightarrow 78$
 $\boxed{95}$



$$A = a + b + d + e$$

$$A \cup B = a + b + c + d + e + f$$

$$A \cap B = b + d$$

$$A \cup B \cup C = a + b + c + d + e + f + g$$

$$A \cap B \cap C = e$$

$$A \cup B \cup C = A + B + C$$

$$- A \cap B - B \cap C - C \cap A$$

$$+ A \cap B \cap C$$

Example A:

$\boxed{08} \rightarrow$ Inclusive / Exclusive
 $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$

Out of 100 students in a school, 60 students play cricket and 52 students play football. There are 5 students who play neither of these games.

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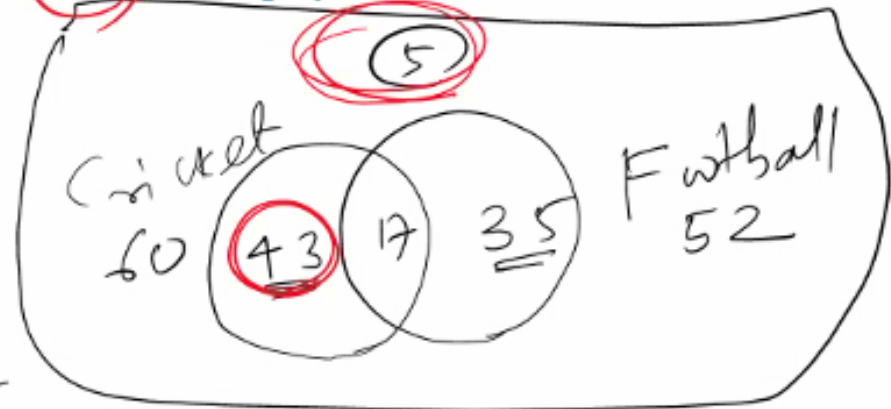
3) How many students play only football? $\rightarrow 35$

4) How many students play either cricket or football but not both? $\rightarrow 78$

5) How many students play exactly one game?

6) How many students play cricket or football?

7) How many students don't play football?



$$(43 + 35)$$
$$(95 - 17)$$

$$\boxed{95}$$

$$43 + 5 = 48$$
$$100 - 52 = 48$$

Example B:

Out of 4000 students in a College, 1500 watch Amazon Prime, 2000 watch Netflix, and 2500 watch HOT STAR. Also 500 watch both Amazon Prime and Netflix, 800 watch both Amazon Prime and HOT STAR and 1100 watch both Netflix and HOT STAR.

I

1. How many watch all?

- (a) 4000
- (b) 2400
- (c) 400
- (d) Cannot be determined

Example B: $A \cup B \cup C$

Out of 4000 students in a College, 1500 watch Amazon Prime, 2000 watch Netflix, and 2500 watch HOT STAR. Also 500 watch both Amazon Prime and Netflix, 800 watch both Amazon Prime and HOT STAR and 1100 watch both Netflix and HOT STAR.

1. How many watch all?

(a) 4000

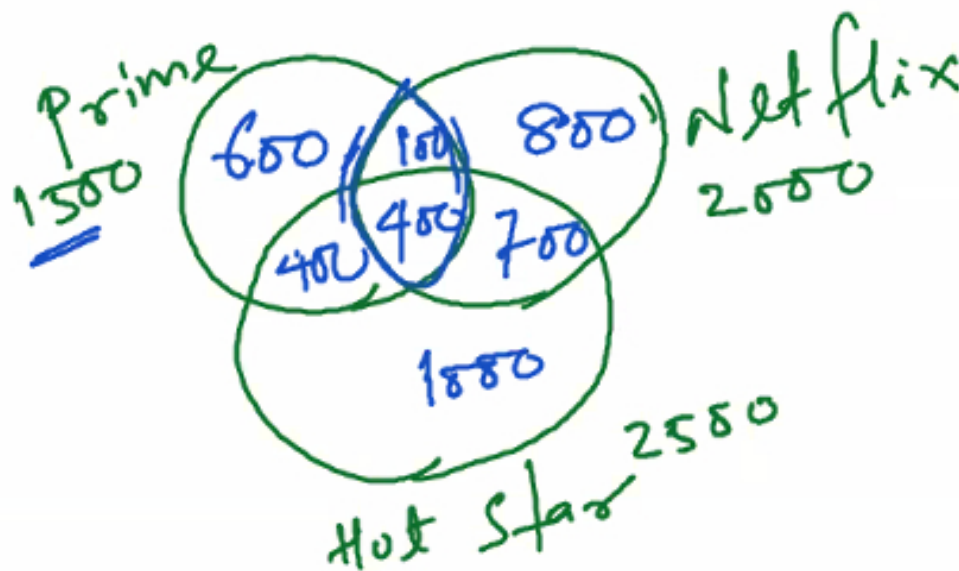
(b) 2400

(c) 400

(d) Cannot be determined

$$A \cup B \cup C = A + B + C - A \cap B - A \cap C - B \cap C + A \cap B \cap C$$

$\cancel{4000}$ $\cancel{1500}$ $\cancel{2000}$ $\cancel{2500}$ $- 500 - 800 - 1100 + 500$



Example 1 $A \cup B \cup C$

A

B

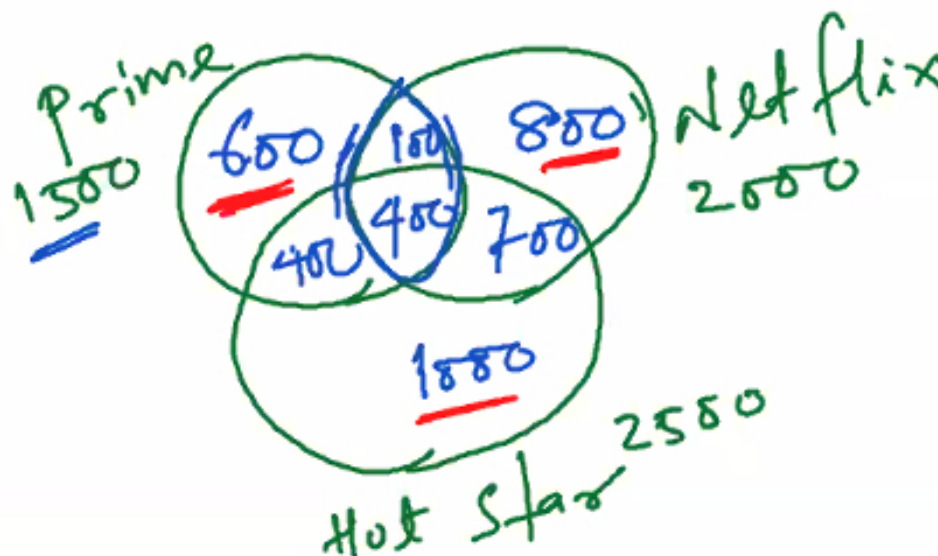
C

Out of ~~4000~~ students in a College, 1500 watch Amazon Prime, 2000 watch Netflix, and 2500 watch HOT STAR. Also 500 watch both Amazon Prime and Netflix, 800 watch both Amazon Prime and HOT STAR, and 1100 watch both Netflix and HOT STAR.

$$A \cup B \cup C = A + B + C - A \cap B - A \cap C - B \cap C + A \cap B \cap C$$

3. How many watch Amazon Prime or HOT STAR?

- (a) 3600
- (b) 3200
- (c) 800
- (d) 2400



Example C:

In a survey of 250 people, it was observed that 15 don't take any of the drinks Tea, Coffee or Juice. 95 take Tea, 120 take coffee and 115 take Juice. 30 take Tea and coffee, 40 take coffee and Juice and 35 take tea and Juice.

1) The number of people who take all the three drinks is:

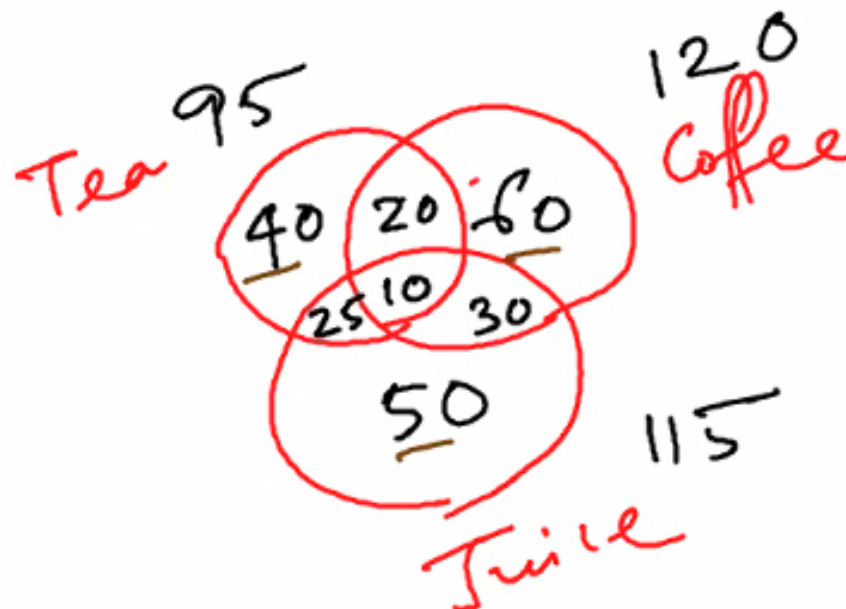
- a) 10**
- b) 15**
- c) 20**
- d) 12**

Example C:

In a survey of 250 people, it was observed that 15 don't take any of the drinks Tea, Coffee or Juice. 95 take Tea, 120 take coffee and 115 take Juice. 30 take Tea and coffee, 40 take coffee and Juice and 35 take tea and Juice.

2) The number of people who take exactly one drink is:

- a) 130
- b) 160
- c) 180
- d) 150

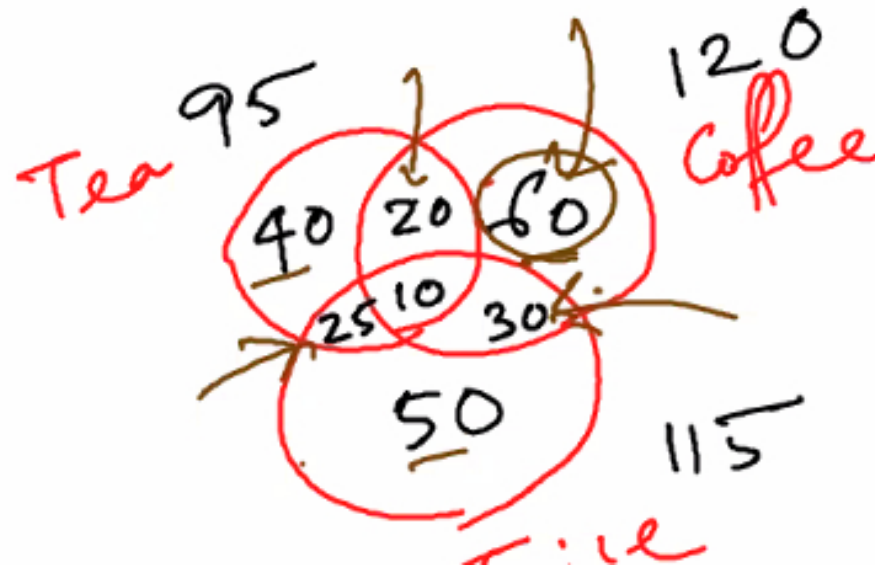


Example C:

In a survey of 250 people, it was observed that 15 don't take any of the drinks Tea, Coffee or Juice. 95 take Tea, 120 take coffee and 115 take Juice. 30 take Tea and coffee, 40 take coffee and Juice and 35 take tea and Juice.

5) The number of people taking Tea & Juice but not Coffee is:

- a) 35
- b) 25
- c) 20
- d) 30



Example D:

A newspaper agent delivers morning newspapers to 300 families. 100 families buy only The Times of India. 25 families buy The Times of India and The Financial Express. 120 families buy The Financial Express. 25 families buy The Financial Express and The Hindu. 25 families buy only The Hindu and no family buy all three newspapers. Find the following:

1. How many families buy The Times of India?
2. How many families buy The Financial Express only?
3. How many families read only one newspaper?
4. Which newspaper is the most popular?
5. Which newspaper is the least popular?

Example E:

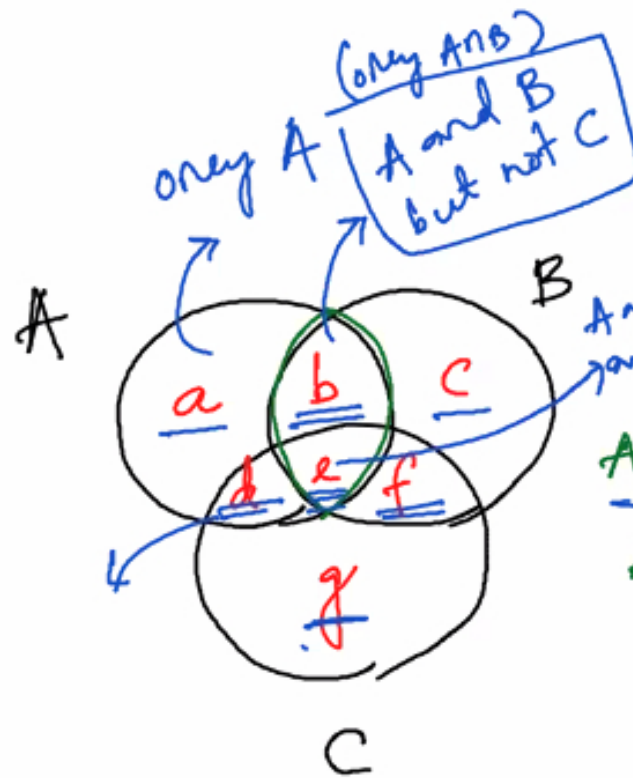
Out of several passengers travelling in a bus, 15 can speak Odia, 6 can speak Bengali and 6 can speak Hindi. There are exactly 2 passengers who can speak exactly 2 languages and only one passenger can speak all the 3 languages. How many passengers are travelling in the bus?

(Assume that any passenger in the bus can speak at least one of the given languages and no passenger in the bus knows any language other than these mentioned languages.)

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$$A \cap B = b + e$$

$$A \cup B \cup C = a + b + c + d + e + f + g$$

$$A \cap B \cap C = e$$

$$\begin{aligned} A \cap B &= b + e \\ B \cap C &= f + e \\ A \cap C &= d + e \end{aligned}$$

$$(b + d + f) + 3e$$

$$A \cup B \cup C = A + B + C$$

$$- A \cap B - B \cap C - C \cap A$$

$$+ A \cap B \cap C$$

$$A \cap B + B \cap C + C \cap A = ?$$

$$A \cup B \cup C$$

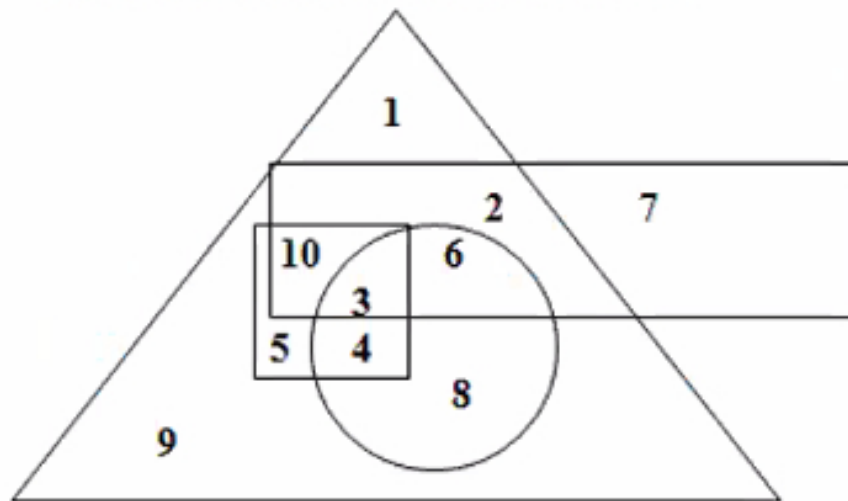
$$= \begin{array}{r} A + B + C \\ \underline{15 \quad 6 \quad 6} \\ - (A \cap B + B \cap C + C \cap A) \end{array} \quad 5$$

$$+ A \cap B \cap C \quad = \quad 23$$

Example F:

The picture given below represents employee structure of an organization.

The rectangle represents technical executives.
The triangle represents marketing executives.
The circle represents production engineers.
The square represents HR managers.



1. Which of the following statements is true?

- a) All technical executives are marketing executives.
- b) Some production engineers are HR managers as well as technical executives.
- c) All marketing executives are production engineers.
- d) Area 2 represents only marketing executives.